# Functions & Proportionality

## Lesson 2 / Overview

Students will graph functions and make connections between the rule and graph as well as between the patterns in the table and the graph. Students will label graphs as linear or quadratic and be able to determine from the rule, table or graph whether that representation models a linear or quadratic function.

**Materials**
**POD:** Overhead POD 2
**Student Pages:**
Graphing Funct; TI-73 Quick Ref Sheet; Pick's Theorem; Peg Game;
**Other Materials:**
Graph paper, graphing calculators, Peg games for each group; transparency of Graph Functions y = 2x + 2

## Teaching Actions

**POD:** *What is the 20th term in the sequence: 2, 6, 12, 20, 30...* This sequence grows by 4, 6, 8, 10... Students can solve the problem by extending the sequence using this pattern. Since this growth shows that the function rule is quadratic students may find the rule: $y = n^2 + n$ to solve the problem. The 20th term is: 420.

**Review:** Go through students' comments from previous day's Big Idea forms. Look for consensus among forms on the big ideas which were developed in Day 1.

### *Large Group Introduction*

1. Show the left-hand side of the overhead **Graphing Functions: y = 2x + 2**. Go through the steps for graphing a function with paper and pencil and have students graph y = 2x + 2.

- Build a table of points
- Get grid paper
- Draw a coordinate axis
- Determine interval for x and y
- Plot points; connect

2. Ask: Is it always necessary for the x and y axis to be labeled in the same manner? Under what conditions would different scales make sense?

## Comments

1. Two functions examined so far are linear and quadratic. Linear functions (y = mx + b) involve the four basic operations (+, -, x, ÷). Quadratic functions include the term, $x^2$.

2. The big idea for this lesson is to reinforce differences between linear and quadratic functions written algebraically and their corresponding graphs and tables.

3. Important differences between tables of each function type is the presence of a constant difference.

Constant differences are looked for across the dependent variable data.

If the first difference is a constant then the function is linear [ graph is a straight line] and is in the form y = mx + b; if the second difference is a constant then the function is quadratic [graph is a parabola] and is in the form of $y = ax^2 + bx + c$.


**© Kathleen Cramer 2000**

## Teaching Actions

3. Now move to the right side of the transparency and go through the steps to graph the function using the graphing calculator.

- Enter function using **y =**
- Adjust **window** to set the scales for x and y axis
- Use **graph** key to show the graph
- Use the **trace** key to find other values for the table

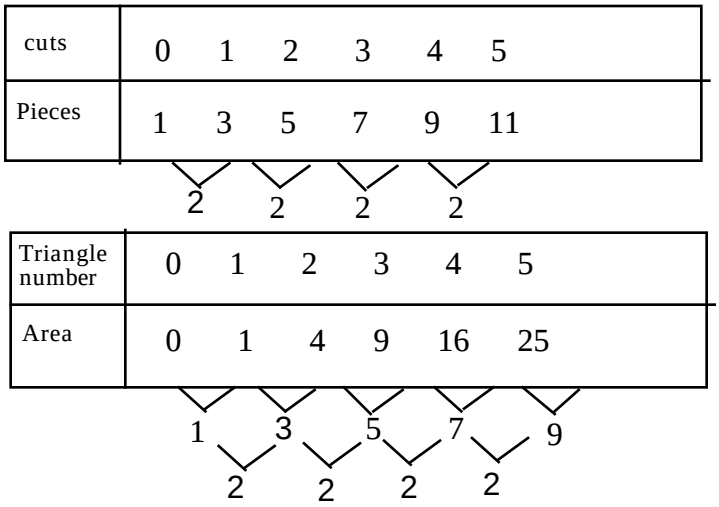
[Have students record on Quick Reference Sheet what each key does; the bottom of the sheet will be completed in the next lesson].

### *Small Group Work*

4. Ask students to use their graphing calculators to graph these functions from Day 1: **p = 2c + 1** and **A = $N^2$.**

### *Large Group Discussion*

5. Compare patterns in tables, rules and graphs for two functions from Day 1: **p = 2c + 1** and **A = $n^2$**.




6. Ask students to restate the connection between patterns in the tables and rules. Compare the rules to the graphs. Ask: What are the differences? How does the curve ( or line) relate to how fast the dependent variable increases? Introduce term: parabola. Compare patterns in tables and graph.

## Comments

4. Course uses TI-73 graphing calculators.

5. You should note that when students graph the functions from day 1 the graphs do not exactly match the realities of the data. For example, with the string activity the data point (2.5, 6) does not match the real context. The graphs on the calculator represent the mathematical world.

6. Emphasize how the dependent variable grows as the independent variable increases.

- linear: steady, constant growth;
- quadratic: more rapid growth,



**© Kathleen Cramer 2000**

## Teaching Actions

7. Label the functions for the straight line graphs as linear functions; label the graphs for parabolas as quadratic functions.

8. Stop and ask students to write in their notes all the connections they see between tables, graphs and algebraic rules they have explored.

***Break***

### *Large Group Instruction*

9. Set the stage for a new exploration by reviewing how to find area on a geoboard.

10. Start the exploration for Pick's Theorem by asking each group to construct a figure with 6 pegs touching the rubber band [on the boundary] and one inside; then have them find the area. Observe that all the areas are the same.

11. Repeat for 3 pegs on the boundary and 0 pegs inside.

### *Small Group Work*

12. State the question: I wonder what relationship exists between boundary points, inside points and area? Use **Pick's Theorem** activity to answer that question.

13. Use graphing calculators to graph each function.

14. As students finish assign **Peg Game** activity.

## Comments

7. When students have difficulty finding the rules, they can rely on connections among tables, rules and graphs to help them.

For example, if a student is having difficulty finding the rule for the Peg Game, encourage the student look a the table for constant differences or graph the data points.

With this information, students shou see that the rule is quadratic; student then can test out some simple rules involving $x^2$.



**© Kathleen Cramer 2000**

## Teaching Actions

### *Large Group Discussion*

15. Share results of Pick's Theorem activity. Record individual rules and then construct the general rule.

Rules for Pick; Theorem:
A = 1/2 P -1
A = 1/2 P
A = 1/2 P +1
A = 1/2 P + 2
**A = 1/2 P + (I -1)**

16. Ask: What type of functions are each of the rules in Pick's theorem? In what ways can you tell that they are linear?

17. Look at the constant differences in the table for the Peg Game. The constant differences occurs as the "difference of the differences". Ask: What type of function must this be? What will the graph look like?

18. The Peg Game rule is: $y = x^2 + 2x$.

19. Summarize information on **Graphing Functions** sheet.

20. Extra practice: **Practice Building Tables and Graphing**.

21. End class by having groups complete **Big Idea** forms.

## Comments

8. Peg Game Table

| # pairs of pegs | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| # moves | 3 | 8 | 15 | 24 |

5 7 9

2 2

9. Students may need help checking for differences with tables involving negative values.



**© Kathleen Cramer 2000**

Transparency

# Graphing Functions : y = 2x + 2

| Using Paper & Pencil | Using TI-82 |
|---|---|
| 1. | 1. Enter function using: y = |
| 2. | 2. Adjust window to set intervals for x-axis and y-axis; set range for values across x-axis and y-axis. |
| 3. | 3. Graph |
| 4. | 4. Use trace key to find other value for tables |
| 5. | |



**© Kathleen Cramer 2000**

# Graphing Functions

**String Activity:**

Condition 1: $y = 2x + 1$

Condition 2: $y = x + 2$

Condition 3: $y = 2x + 2$

**Patterns with Squares [perimeter]:**

$y = 4x$

**Description of graphs:**

**Patterns with squares [area]:**

$y = (x^2 + x) / 2$

**Equilateral triangles:**

$y = x^2$

**Description of graphs:**



**© Kathleen Cramer 2000**

# TI-73 Quick Reference Sheet

| KEY | What it does/how to use |
|---|---|
| y = | |
| x | |
| Window | |
| Graph | |
| Trace | |
| 2nd | |
| **TbleSet** Window | |
| **Table** Graph | |



**© Kathleen Cramer 2000**

# Pick's Theorem

## Discovering Pick's Theorem

**Task 1:** Explore the patterns involving area of shapes on the geoboard with 0 pegs inside a polygon. Collect data for this table:

| # of pegs on the boundary | 3 | 4 | 5 | 6 | 7 |
|---|---|---|---|---|---|
| Area | | | | | |

It is possible to determine area of a polygon with 0 pegs inside the polygon if you know the number of pegs touching the boundary. If a shape has 12 pegs touching the boundary and 0 pegs inside, what is its area? Use patterns in the table to determine the area. Describe the patterns.

Write a rule to show the relationship between the number of pegs touching the boundary and area when there are 0 pegs inside:

Area =_____

Test out your rule for all the number pairs in the table.

Use the rule to determine the area of a shape with 17 pegs touching the boundary with 0 pegs inside.



**© Kathleen Cramer 2000**

**Task 2:** Continue with your explorations. Vary the number of pegs inside the figure to see how the relationship between area and pegs touching the boundary changes.

1 peg inside

| # of pegs on boundary | Area |
|---|---|
| | |

2 pegs inside

| # of pegs on boundary | Area |
|---|---|
| | |

3 pegs inside

| # of pegs on boundary | Area |
|---|---|
| | |

Describe patterns observed in each table.

What would the area be if there were 12 pegs on the boundary and 1, 2 and then 3 pegs inside?

Write a rule to explain the relationship between area and pegs on the boundary for each case above.



**© Kathleen Cramer 2000**

**EXTENSION:** Pick's theorem explains the relationship among 3 variables: Number of pegs on the boundary, Number of pegs inside the figure and area.

**Let A = area** **Let B = Pegs on the boundary** **Let I = Pegs inside**

Write a rule that can be used to find the area of a shape on the geoboard given "B" pegs on the boundary and "I" pegs inside the boundary.

Test out your rule with data from the four tables.

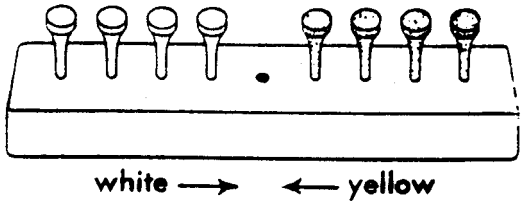
Explain how you thought through this problem.



**© Kathleen Cramer 2000**

# The Peg Game

The object of this puzzle is to interchange the yellow and white pegs. You must move the pegs according to the following rules:




1. The white pegs must move only to the right; the yellow pegs must move only to the left.
2. You can move only one peg at a time.
3. You can move a peg into an adjacent hole.
4. You can jump, but only a single peg of the opposite color.

**Task 1:** Take a few minutes trying to solve the puzzle. Record the number of moves for each attempt.

**Task 2**: Can you do the puzzle with 4 pegs? Record the number of moves for each attempt. Is there a way to tell whether you have the minimum number of moves or not?


**© Kathleen Cramer 2000**

**Task 3:** Try the puzzle using 1 pair of pegs, 2 pairs, 3 pairs and 4 pairs. Is there a relationship between the number of pairs of pegs and the corresponding minimum number of moves? Collect the data and record on the table below:

| # of pairs of pegs [x] | 1 | 2 | 3 | 4 | |
|---|---|---|---|---|---|
| # of moves [y] | | | | | |

**Task 4:** Can you figure out a rule that works for all the number pairs so you can predict the number of moves given the number of pairs of pegs?

Check the rule for data set.

**Task 5:** Graph the data on a coordinate axis. Describe the graph

**Task 6:** If you started with 15 pairs of pegs and it took 1 second for each move, about how long would it take to transfer the pile?


**© Kathleen Cramer 2000**

# Practice Building Tables and Graphs

I. Build a table of 5 sample points that fit each rule. Graph each rule.

**$y = 3x^2 - 1$**

| x | y |
|---|---|
| -2 | |
| -1 | |
| 0 | |
| 1 | |
| 2 | |

**$y = x + 2$**

| x | y |
|---|---|
| -3 | |
| -2 | |
| -1 | |
| 0 | |
| 1 | |

**$y = x^2 - 3$**

| x | y |
|---|---|
| | |

**$y = 3x + 2$**

| x | y |
|---|---|
| | |

**Look at the constant differences in each table. What is true for quadratic functions? What is true for linear functions?**



**© Kathleen Cramer 2000**